

Your Paper Title Here

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Date

Abstract

Write your abstract here.

Keywords

Keyword 1, Keyword 2, Keyword 3, Keyword 4

Introduction / Background

In December 2020, Pfizer and BioNTech began distributing their two dose vaccine for COVID-19.

The data comes from blah blah blah

make a segmented barplot and table of data (side by side)

From the initial data we have these statistics.

```
## Warning: package 'knitr' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

Table 1: Table 1: Vaccine Efficacy against Covid-19 at least 7 days after second dose in patients without evidence of infection

Group	Cases	Sample_size
BNT162b2	8	17411
Placebo	162	17511
Total	170	34922

Table 2: Estimated Risk and Vaccine Efficacy

Statistic	Formula	Estimate
$\hat{\pi}_v$ (vaccine infection probability)	$8/17411$	0.000459
$\hat{\pi}_p$ (placebo infection probability)	$162/17511$	0.009251
RR (relative risk)	$\hat{\pi}_v/\hat{\pi}_p$	0.049666
ψ (vaccine efficacy)	$1 - \hat{\pi}_v/\hat{\pi}_p$	0.950334



Statistical Methods

Model

Instead of looking at two independent binomial models, since the trial ran until a predetermined 170 observations of COVID-19 infections across both groups was observed, we can use a single binomial random variable as our statistical model (refer to appendix for proof). We can denote random variable T as the number in the vaccine group from the total 170 observations of infection. T can be modeled as (refer to appendix for proof):

$$T \approx \text{Binom}(n = 170, \pi)$$

Here $t_{obs} = 8$ and $\pi = P(\text{case occurred in vaccine group})$. We can write π as $\pi = \frac{\pi_v n_v}{\pi_v n_v + \pi_p n_p} = \frac{\pi_v}{\pi_v + \pi_p}$ since the sample sizes of the group receiving the placebo and the group receiving the vaccine are

roughly the same ($n_p \approx n_v$).

Our parameter of interest is vaccine efficacy. We can rewrite this, ψ , in terms of π to get (refer to appendix for proof):

$$\psi = \frac{1 - 2\pi}{1 - \pi} \quad -\infty < \psi < 1$$

With this we can also find π in terms of ψ for later use in our Frequentist and Bayesian analysis of our data. We get (refer to appendix for proof):

$$\pi = \frac{1 - \psi}{2 - \psi} \quad 0 < \pi < 1$$

Because the FDA requires at least 30% efficacy for a new therapy to be approved, the hypothesis we are interested in testing for both approaches is

$$H_0 : \psi = 0.3 \quad H_1 : \psi > 0.3$$

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

For the Bayesian approach, we are using a Beta binomial model where

$$T|\pi \sim \text{Binom}(n = 170, \pi)$$

π is assumed to follow a Beta distribution with shape parameters a and b :

$$\pi \sim \text{Beta}(a, b)$$

As stated earlier, our parameter of interest is vaccine efficacy represented by $\psi = \frac{1-2\pi}{1-\pi}$

Prior Distribution

To determine the prior distribution of π , a $Unif(0, 1)$ distribution which corresponds to $a = 1$ and $b = 1$ can be used as a non-informative prior. However, we chose to use a prior based on beliefs about vaccine efficacy, ψ . Those were:

$$P(\psi \geq 0.3) = 0.01$$

$$P(\psi \geq 0) = 0.5$$

These mean that there is a 1% chance that the vaccine is more than 30% or higher and that there is a 50% chance that the vaccine works. We chose a pessimistic prior to help ensure vaccine efficacy is not overstated after statistical analysis.

The probability statements can be rewritten in terms of π , since π 's distribution is known:

$$P(\psi = \frac{1 - 2\pi}{1 - \pi} \geq 0.3) = 0.01$$

$$P(\pi \leq \frac{7}{17}) = 0.01$$

$$P(\psi = \frac{1 - 2\pi}{1 - \pi} \geq 0) = 0.5$$

$$P(\pi \leq \frac{1}{2}) = 0.5$$

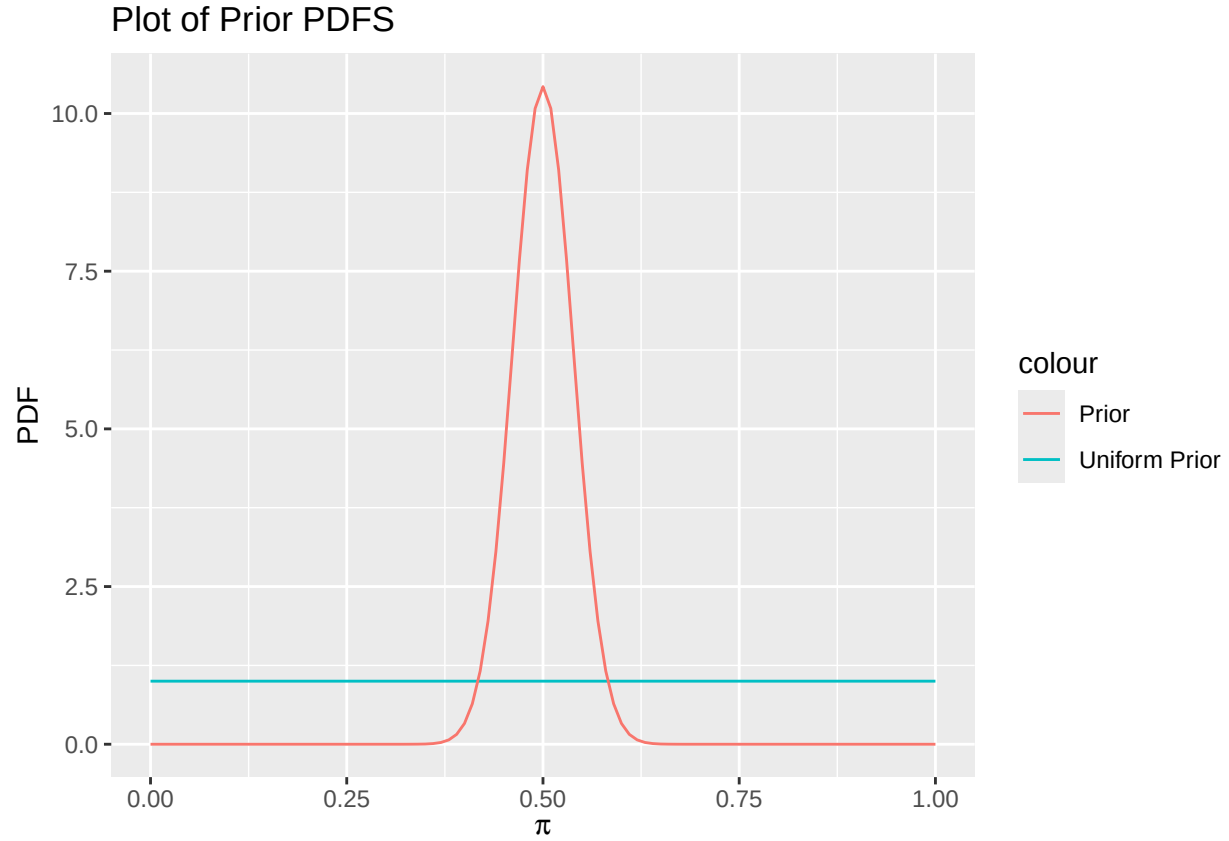
This means that the $Beta(a, b)$ prior for π has a median of 0.5 and 1st percentile of $\frac{7}{17}$. These quantiles can be plugged into R using `beta.select()` to select the shape of the distribution.

```
## [1] 85.63 85.63
```

The prior distribution for π is:

$$\pi \sim Beta(85.63, 85.63)$$

The distribution is shown below, along with what the uniform prior looks like:



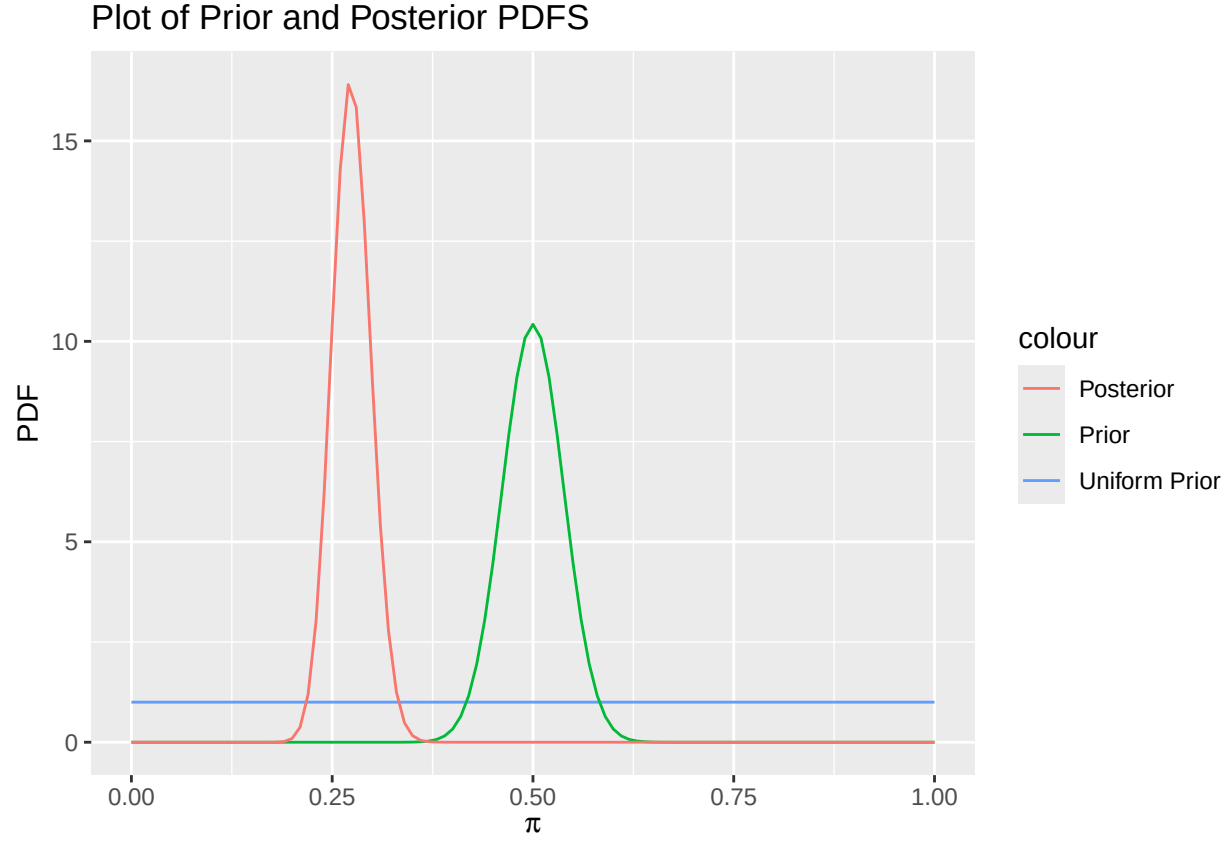
Posterior Distribution

The posterior distribution can be determined with its shape parameters being $a + t$ and $b + n - t$, where t is the observed value for T . In this case, $a = 85.63$, $b = 85.63$, $n = 170$, and $t = 8$. So:

$$\pi|t \sim \text{Beta}(85.63 + 8, 85.63 + 170 - 8)$$

$$\pi|t \sim \text{Beta}(93.53, 247.53)$$

The posterior can be plotted in comparison to the prior to determine if it is skewed.



Inference The median and interval of ψ , vaccine efficacy, can be found in relation to the posterior distribution of π .

$$P\left(\frac{1-\psi}{2-\psi} < m|t\right) = 0.5$$

This equation can be solved for ψ for the median and for the interval values by using corresponding quantile values.

For hypothesis testing, the null hypothesis, as stated earlier is:

$$H_0 : \psi = 0.3$$

with an alternative that vaccine efficacy is higher than 30%:

$$H_1 : \psi > 0.3$$

The posterior probability is represented by

$$P(\psi \leq 0.3|t)$$

This can be converted in terms of π to make the calculation possible since π 's distribution is known:

$$P(\frac{1-2\pi}{1-\pi} \leq 0.3|t)$$

This corresponds to finding the following probability:

$$P(\pi \geq \frac{7}{17}|t)$$

Results

Using Bayesian inference, the median of π was calculated to be 0.2738. For ψ , the corresponding median value is 0.6227. Since the posterior PDF is right skewed in relation to the prior the Highest Posterior Density Interval was used to determine the interval. For π , the 95% HPDI is (0.2274, 0.3227). Using these values to solve the 2.5th and 97.5th percentiles of ψ in terms of π , the HPDI for ψ is (0.5261, 0.7057). Under the posterior distribution, the probability of vaccine efficacy being less than or equal to 30% was found to be $5.6848 \cdot 10^{-08}$.

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from <https://www.example.com>.

Appendix

Code

```
beta.select(quantile1 = list(p = 0.01, x = 7/17),  
            quantile2 = list(p = 0.5, x = 0.5))
```

```
ggplot() +  
  geom_function(fun = dbeta,  
               mapping = aes(color = "Uniform Prior"),  
               args = list(shape1 = 1, shape2 = 1),  
               xlim = c(0, 1)) +  
  geom_function(fun = dbeta,  
               mapping = aes(color = "Prior"),  
               args = list(shape1 = 85.63, shape2 = 85.63),  
               xlim = c(0, 1)) +  
  labs(  
    title = "Plot of Prior PDFS",  
    x = expression(pi),  
    y = "PDF")
```

```
ggplot() +  
  geom_function(fun = dbeta,  
               mapping = aes(color = "Uniform Prior"),  
               args = list(shape1 = 1, 1),  
               xlim = c(0, 1)) +  
  geom_function(fun = dbeta,
```

```

    mapping = aes(color = "Prior"),
    args = list(shape1 = 85.63, 85.63),
    xlim = c(0, 1)) +
geom_function(fun = dbeta,
    mapping = aes(color = "Posterior"),
    args = list(shape1 = 93.53, 247.53),
    xlim = c(0, 1)) +

labs(
  title = "Plot of Prior and Posterior PDFS",
  x = expression(pi),
  y = "PDF"
)

```

Proofs

Proof that $T \approx \text{Binom}$: We start with our two independent binomial models $X \sim \text{Binom}(n_v, \pi_v)$ and $Y \sim \text{Binom}(n_p, \pi_p)$. Here, X is the number of COVID-19 cases in the vaccine group and Y is the number of COVID-19 cases in the placebo group. Restricting our attention to only the COVID-19 cases, we can define the random variable T as the number of COVID-19 cases in the vaccine group given that the total number of COVID-19 cases in both groups is 170. In other words, T is the value of X given $X + Y = 170$.

So, we can find the PMF of T by:

$$\begin{aligned}
 P(T = t) &= P(X = t | X + Y = n) \quad t = 0, 1, \dots, n \\
 &= \frac{P(X = t \cap X + Y = n)}{P(X + Y = n)} \\
 &= \frac{P(X = t \cap Y = n - t)}{P(X + Y = n)}
 \end{aligned}$$

Since X and Y are independent,

$$= \frac{P(X = t) \times P(Y = n - t)}{P(X + Y = n)}$$

We'll come back to this, but for now, referring back to table 2 in the introduction, we can see that n_v and n_p are very large and π_v and π_p are very small. So, these two binomial models can be approximated by 2 independent poisson distribution: $X \approx Pois(\lambda_v)$ and $Y \approx Pois(\lambda_p)$ where $\lambda_v = n_v \pi_v$ and $\lambda_p = n_p \pi_p$.

For a poisson distribution $P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$ So,

$$P(X = t) = \frac{\lambda_v^t e^{-\lambda_v}}{t!} \quad P(Y = n - t) = \frac{\lambda_p^{n-t} e^{-\lambda_p}}{(n-t)!}$$

Since the two poisson distributions are still independent of each other, we can add them to get $X + Y \sim Pois(\lambda_v + \lambda_p)$. So,

$$P(X + Y = n) = \frac{(\lambda_p + \lambda_v)^n e^{-\lambda_p - \lambda_v}}{n!}$$

Plugging these into the PMF we found for T , we get:

$$P(T = t) = \frac{\frac{\lambda_v^t e^{-\lambda_v}}{t!} \times \frac{\lambda_p^{n-t} e^{-\lambda_p}}{(n-t)!}}{\frac{(\lambda_p + \lambda_v)^n e^{-\lambda_p - \lambda_v}}{n!}}$$

$$P(T = t) = \frac{n!}{t!(n-t)!} \times \frac{\lambda_v^t \lambda_p^{n-t} e^{-\lambda_p} e^{-\lambda_v}}{(\lambda_p + \lambda_v)^n e^{-\lambda_p - \lambda_v}}$$

$$P(T = t) = \frac{n!}{t!(n-t)!} \times \frac{\lambda_v^t \lambda_p^{n-t}}{(\lambda_p + \lambda_v)^n}$$

$$P(T = t) = \frac{n!}{t!(n-t)!} \times \frac{\lambda_v^t \lambda_p^{n-t}}{(\lambda_p + \lambda_v)^t (\lambda_p + \lambda_v)^{n-t}}$$

$$P(T = t) = \frac{n!}{t!(n-t)!} \times \frac{\lambda_v^t}{(\lambda_p + \lambda_v)^t} \times \frac{\lambda_p^{n-t}}{(\lambda_p + \lambda_v)^{n-t}}$$

The choose function: $\frac{n!}{t!(n-t)!} = \binom{n}{t}$

$$P(T = t) = \binom{n}{t} \times \left(\frac{\lambda_v}{(\lambda_p + \lambda_v)}\right)^t \times \left(\frac{\lambda_p}{(\lambda_p + \lambda_v)}\right)^{n-t}$$

We can rewrite the third term $\left(\frac{\lambda_p}{(\lambda_p + \lambda_v)}\right)^{n-t}$ as

$$\left(1 - \frac{\lambda_v}{(\lambda_p + \lambda_v)}\right)^{n-t}$$

$$P(T = t) = \binom{n}{t} \times \left(\frac{\lambda_v}{(\lambda_p + \lambda_v)}\right)^t \times \left(1 - \frac{\lambda_v}{(\lambda_p + \lambda_v)}\right)^{n-t}$$

And yay! Looking at this closely, this is the binomial PMF for $Binom(n, \frac{\lambda_v}{(\lambda_p + \lambda_v)})$. So, we can say

$$T \approx Binom(n, \frac{\lambda_v}{(\lambda_p + \lambda_v)})$$

since $\lambda_v = n_v \pi_v$ and $\lambda_p = n_p \pi_p$, we can say

$$T \approx Binom(n, \frac{n_v \pi_v}{n_p \pi_p + n_v \pi_v}) \quad \text{where } n = 170$$

Finally, knowing that $n = 170$ and since n_v and n_p are really large, we can say for $\pi = \frac{\pi_v}{\pi_p + \pi_v}$

$$T \approx Binom(170, \pi)$$

Proof of $\psi = \frac{1-2\pi}{1-\pi}$: The definition of vaccine efficacy (ψ) is: $\psi = 1 - \frac{\pi_v}{\pi_p}$, we can rewrite this in terms of π using:

$$\pi = \frac{\pi_v}{\pi_v + \pi_p}$$

$$\pi(\pi_v + \pi_p) = \pi_v$$

$$\pi \pi_v + \pi \pi_p = \pi_v$$

$$\pi \pi_p = \pi_v - \pi \pi_v$$

$$\pi\pi_p = \pi_v(1 - \pi)$$

$$\frac{\pi_v}{\pi_p} = \frac{\pi}{1 - \pi}$$

Plugging $\frac{\pi_v}{\pi_p} = \frac{\pi}{1 - \pi}$ into $\psi = 1 - \frac{\pi_v}{\pi_p}$, we get:

$$\psi = 1 - \frac{\pi}{1 - \pi}$$

$$\psi = \frac{1 - \pi}{1 - \pi} - \frac{\pi}{1 - \pi}$$

$$\boxed{\psi = \frac{1 - 2\pi}{1 - \pi} \quad -\infty < \psi < 1}$$

Proof of $\pi = \frac{1-\psi}{2-\psi}$: Using the equation we deduced in the previous proof we can rearrange $\psi = \frac{1-2\pi}{1-\pi}$ to create an equation for π in terms of ψ .

$$\psi = \frac{1 - 2\pi}{1 - \pi}$$

$$\psi(1 - \pi) = 1 - 2\pi$$

$$\psi - \psi\pi = 1 - 2\pi$$

$$2\pi - \psi\pi = 1 - \psi$$

$$\pi(2 - \psi) = 1 - \psi$$

We finally get

$$\boxed{\pi = \frac{1 - \psi}{2 - \psi} \quad 0 < \pi < 1}$$